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# An alternative method to IEC 60891 standard for I − V curve correction based on a single measurement

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ABSTRACT

In the expanding field of photovoltaic solar energy, it is often necessary to apply a correction method to I-V curves measured outdoors and translate them to standard test conditions (STC). The international standard IEC 60891 serves as a primary reference for correction methods, but its application is not straightforward in the case of outdoor photovoltaic strings. Even for photovoltaic modules, this standard requires measurements of several I-V curves under controlled conditions to determine the parameters, which makes the procedure experimentally complex. An essential correction method should avoid additional measurements and provide an uncertainty comparable to IEC 60891, especially for outdoor measurements. In this paper, a novel method is presented in which the correction of the I-V curve is performed without additional measurements. The method is based on procedure 1 of IEC 60891, in which the parameters for the correction of the I-V curve are determined by iterative processes via the linear correction of the maximum power with temperature and irradiance. The new method was tested for both indoor and outdoor measurements. The method showed an uncertainty of less than ± 2 % in the maximum power of the corrected curves. This method can be implemented in spreadsheets and offers an alternative with similar accuracy to method 1 of IEC 60891, while maintaining the simplicity of the correction by using only the measured curve and the values of the temperature coefficients of the photovoltaic modules.

## 1. Introduction

The use of photovoltaic (PV) systems to generate electricity has increased considerably over the last two decades. Since it is an intermittent energy source, the power output of photovoltaic modules varies over time and depends mainly on the solar irradiance on the PV module ($G$) and the operating temperature of the PV cell ($T_c$). The prediction of the power output of a PV system over time is based on mathematical models where the input parameters are usually provided under standard test conditions (STC), where $G = 1000$ W/m$^2$ with a standard spectrum AM 1.5 and $T_c = 25$ °C. The measurement of the current–voltage (I − V) curve is necessary to determine the electrical parameters of PV modules and can be performed indoors or outdoors [1]. The main parameters of interest for a PV module are the maximum power ($P_m$), the open circuit voltage ($V_{oc}$) and the short circuit current ($I_{sc}$). Outdoor I − V curve measurements are rarely performed at STC, which requires correction methods for the measured curve, as the electrical parameters of a PV module are typically required under STC for various practical purposes.

With the increasing number of PV installations over the last decade, both in distributed generation and large-scale PV systems, the measurement of outdoor I − V curves is a fundamental resource for the commissioning, fault diagnosis and maintenance of PV systems. As measurement conditions change over time, a method to correct I − V curves for other conditions of interest is required, with STC being the most commonly used condition. Measurements under STC are also important to establish a reference that can be used in future comparisons, e.g. to determine the degree of degradation of modules, defects and various losses in a PV system.

The value of $P_m$ for a PV string in operation can be determined via the measuring system of the inverter for energy monitoring. When evaluating power generation performance, models with three inputs for maximum power have shown good performance [2]. However, the measurement of I − V curves for outdoor strings and arrays allows the determination of $P_m$ and the diagnosis of anomalies and serves as a more comprehensive tool for analyzing PV systems, and some inverter models even have this feature.

Correcting I − V curve measurements of PV systems to STC or other conditions is a technical challenge when it comes to determining

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parameters such as series resistance ($R_s$), which is usually derived from several measurements under controlled conditions. The international standard IEC 60891 [3] provides guidelines for the correction of I − V curves for different $G$ and $T_c$ conditions; however, the methods described appear to be primarily applicable to individual PV modules. Regarding the measurement of I − V curves for outdoor PV systems, it is not practically feasible to measure I − V curves at different $G$ levels and the same $T_c$ to estimate $R_s$ or other parameters used in interpolation methods. In addition, Procedures 1 and 2 of IEC 60891 require I − V curve measurements at the same $G$ and different $T_c$ to estimate the curve correction factor ($k$).

Since determining the parameters $R_s$ and $k$ using the methods described in IEC 60891 is not easy, these procedures are rarely applied to PV systems without making simplifications that inevitably limit the accuracy of the corrected I − V curve results. Therefore, a method that can correct an I − V curve, whether for a module, a string or an array and that is accurate enough to correct the entire I − V curve, is crucial for the importance of the market that photovoltaic solar energy has already achieved and continues to expand.

The procedures for correcting the I − V curve have been an issue for some time [4], and the first edition of the IEC 60891 standard dates back to 1987 [5]. The translation procedures of I − V curves in the IEC 60891 standard were derived from algebraic procedures [6], and the original equation from the first version of the standard is still included in Procedure 1 of the most updated version [7]. Correction procedures for I-V curves are a current research topic due to the economic and technical importance of their results, and the different methods of IEC 60891 have recently been evaluated and compared [8]. There are several technical challenges in determining the parameters for the application of IEC 60891 Procedures 1 and 2, and Procedure 3 requires at least two I − V curves measured under different conditions [6].

Various methods for translating the I − V curve were evaluated by Piliougine et al. [9] using a small set of experiments. It was found that Procedure 3 of the IEC 60891 standard is a good choice and requires only one interpolation [9]. Preferring simpler methods over complex ones is advisable, as the latter often provide marginal improvements in accuracy. However, even procedures that involve interpolation require more than one measurement of the I − V curve under conditions that are difficult to establish for a PV system. To complete the scenario, the determination of the parameters used in procedures 1 and 2 of IEC 60891 is performed by iterative processes, which leads to errors that can be minimized with optimization tools [10].

The latest version of the IEC 60891 standard [7] presents four procedures for correcting the I − V curve of a PV module to STC. To apply Procedure 1, $R_s$ and $k$ must be determined, which is a challenge for outdoor measurements. A simplified IEC 60891 method was also evaluated by Dubey et al. [11], in which $R_s$ and $k$ were neglected in the calculations due to the difficulties in determining them [11]. This leads to non-negligible intrinsic errors, as $R_s$ has a significant influence on the I − V curve and there is no typical value that can be estimated for both a module and a PV array.

Another issue that arises when applying Procedure 1 of IEC 60891 concerns the method for determining $R_s$. The procedure for determining $R_s$ was investigated and evaluated by comparing different versions of the standard, which led to different results [12]. An improvement suggested by the authors would be to iteratively sweep the $R_s$ values until the ideal value is found, and not only until the $P_m$ deviation is less than 0.5 %, as stated in the standard. Although the authors conclude that this slight increase in the accuracy of $R_s$ would not significantly improve $P_m$, it is a simple procedure that can be implemented in software [12]. However, determining $R_S$ and $k$ requires relatively complex experimental procedures, especially for outdoor measurements.

Other alternative procedures for correcting I − V curves have been explored, using a simplified approach of procedure 2 of the IEC 60891 standard, which still requires at least two I − V curves measured under different $G$ and $T_c$ conditions [13]. There are also approaches to perform the translation based on a measurement of the I − V curve and using the single diode model to determine the curve under different $G$ and $T_c$ conditions [14]. Correction procedures for faulty I − V curves based on the IEC 60891 standard are also a current research topic, where the curves were synthesized to avoid uncontrolled uncertainties in experimental procedures [15].

One of the challenges in translating I − V curves is nonlinearity, as various parameters exhibit nonlinear behavior, especially with irradiance and temperature [16–18]. The modification of procedure 2 of the IEC 60891 standard in various aspects, including the logarithmic variation of $V_{oc}$ with irradiance, the irradiance-dependent voltage temperature coefficient and the temperature dependence of $R_s$, can improve the conventional procedures [19]. However, procedures that are highly dependent on multiple measurements and complex may affect their applicability in the practical engineering and analysis of PV systems. Any method for correcting the I − V curve should ideally correct the entire curve. Alternative methods that only determine the points of interest are also an option. The correction method SIDT (Standard Irradiance Desired Temperature) has surpassed the simplified procedure 1 of IEC 60891 [20]. However, the correction does not cover the entire I − V curve as $R_s$, and $k$ are kept at zero and the resulting errors for $P_m$ were around ± 2.3 % [20].

The estimated uncertainty of the correction procedure is a factor that should be considered. For example, the uncertainty for the translation of the I − V curve can be evaluated based on synthetic curves and Monte Carlo simulations covering a wide range of $G$ and $T_c$ [21]. There is a recognized need for a simple and easily applicable approach for the correction of a measured I − V curve using IEC 60891 standard procedures, especially for the translation of the I − V curve into STC with uncertainty estimates based on experimental results. When correcting an I − V curve to obtain $P_m$ at STC, it should also be noted that the temperature coefficients obtained from the PV module catalogs themselves represent additional sources of uncertainty [22].

Procedures 1 and 2 of IEC 60891 can provide uncertainties of less than 2 % for the determination of $P_m$, especially in the range of 800 − 1200 W/m$^2$ [23]. Procedures 1 and 2 are well established and retain the simplicity of algebraic operations, apart from the need for additional measurements under controlled conditions. On the other hand, bilinear interpolation methods for translating I − V curves can be even more precise but generally require reference I − V curves under different combinations of $G$ and $T_c$ [24].

Most correction methods that require parameters determined from several measurements are not readily applicable. In this context, this article aims to present a method that has an accuracy comparable to IEC 60891 in correcting the I − V curves of crystalline silicon PV modules and strings to STC and does not require additional measurements beyond the measured I − V curve itself. The method was developed by modifying procedure 1 of IEC 60891 and provides a direct way to translate the I − V curves of PV modules, strings and arrays to a different condition. The irradiance range from 500 to 1000 W/m$^2$ and $T_c$ from 25 to 65 °C was evaluated. To avoid the need to measure several I − V curves to obtain $R_s$ and k for the application of IEC 60891, the linearity of the Pm values for $G$ and $T_c$ is used as a hypothesis. In addition to the measurement conditions, the presented method requires as input only the temperature coefficients of the PV modules, which can be easily found in the technical specifications.

## 2. Methodology

This section first presents the proposed method itself. It then shows how the method was tested and applied using a series of indoor I − V curve measurements, and finally presents the application of the method to an outdoor PV string.

### 2.1. Description of the proposed $I - V$ curve correction method

The method developed for the correction of the I-V curve is based on the hypothesis of a linear variation of $P_m$ with $G$ and $T_c$. Although the linearity is subject to certain limitations within the application range of the method (500 to 1000 W/m$^2$), it allows a simplified implementation. Thus, the input data for the method is summarized in the measurement and correction conditions as well as the temperature coefficients of the PV module. The linear variation of $P_m$ with $G$ and $T_c$ is applied via Eq. (1), used for PV modules, strings or even arrays, which is also found in PVWATTS [25].

$$P_2 = P_1 \cdot \left(\frac{G_2}{G_1}\right)[1 + \gamma(T_2 - T_1)] \tag{1}$$

where: $P_1$, $G_1$ and $T_1$ are respectively the maximum power, irradiance and temperature under the measured condition; $P_2$, $G_2$ and $T_2$ are respectively the maximum power, irradiance and temperature of the target condition; and γ (°C$^{-1}$) is the coefficient of variation of $P_m$ with temperature obtained from the catalog of the module.

The current points of the $I - V$ curve are corrected according to Eq. (2), which can be found in procedure 1 of the IEC 60891 standard.

$$I_2 = I_1 + I_{SC1} \cdot \left[\left(\frac{G_2}{G_1} - 1\right) + \alpha \cdot (T_2 - T_1)\right] \tag{2}$$

where: $I_1$ corresponds to the current values of the $I - V$ pairs from the measured curve, $G_1$ and $T_1$ are respectively the irradiance and the temperature of the measured condition, $I_{sc1}$ is the short-circuit current under the measured condition ($G_1$, $T_1$), α is the coefficient of variation of $I_{sc}$ with temperature (obtained from the module catalog in the unit °C$^{-1}$), $I_2$ corresponds to the current values of the $I - V$ pairs corrected to condition (2), $G_2$ and $T_2$ are respectively the irradiance and the temperature of the corrected condition.

The voltage points of the $I - V$ curve are corrected using Eq. (3) from Procedure 1 of the IEC 60891 standard.

$$V_2 = V_1 - R_S \cdot (I_2 - I_1) - k \cdot I_2 \cdot (T_2 - T_1) + \beta \cdot V_{OC1} \cdot (T_2 - T_1) \tag{3}$$

where: $V_1$ is the voltage values of the $I - V$ pairs from the measured $I - V$ curve, $R_s$ is the series resistance of the PV module or string (to be determined by an iterative method), $k$ is the curve correction factor (to be determined by an iterative method), β is the coefficient of variation of $V_{oc}$ with temperature (to be obtained from the module catalog in the unit °C$^{-1}$), $V_{oc1}$ is the open circuit voltage under the measured conditions ($G_1$, $T_1$) and $V_2$ are the voltage values at target conditions ($G_2$, $T_2$).

The $P_m$ of the I-V curve is calculated by a fourth-degree polynomial fitted in the range of $0.75\ V_{mp} < V < 1.15\ V_{mp}$, as described in the ASTM-1036 M standard [26].

The proposed translation procedure occurs in two stages. In the first, the $I - V$ curve is corrected exclusively with $G$ and then $R_s$ is determined iteratively by the following steps because it has the greatest impact on the initial corrected power calculation:

1a – Calculate the target power $P_2$ considering only the correction with irradiance ($G_2$), i.e. using Eq. (1) with $T_1 = T_2$ to eliminate the temperature correction;

2a – Set the values of $R_s$ and $k$ to zero;

3a – Calculate the current and voltage points for condition $G_2$ using Eq. (2) and (3). To eliminate the temperature correction, set $T_1 = T_2$. Calculate $P_m$ from the $I - V$ curve.

4a – Calculate the difference between the $P_m$ obtained in step 3a and the value obtained in step 1a, i.e. use the value of $P_2$ as the target value for $P_m$ from the I-V curve.

5a – If the difference obtained in step 4a is greater than 0.25 %, add 10 mΩ to $R_s$ and repeat steps 3a and 4a until the specified difference between the target value $P_2$ and $P_m$ from the I-V curve is obtained.

The IEC 60891 standard prescribes a difference of 0.5 % for the convergence of the iterative process to determine $R_s$. In the proposed method, a difference of 0.25 % was used to define the convergence of $P_m$ calculated from the $I - V$ to the target power $P_2$.

Once the condition in step 5a is met, the iterative process to determine $R_s$ is terminated and the new phase begins where the I-V curve is corrected to conditions ($T_2$, $G_2$). The procedure consists of the following steps:

1b – Calculate the target power $P_2$ using Eq. (1) with temperature and irradiance correction, i.e. $T_1$ and $G_1$ are measured values and $T_2$ and $G_2$ are target values;

2b – Calculate the current and voltage points using Eq. (2) and (3) and calculate $P_m$ from the $I - V$ curve. In this step, the $I - V$ curve is calculated using the $R_s$ from the previous step 5a and $k = 0$;

3b – Calculate the difference between $P_m$ from the I-V curve determined in step 2b and the target power ($P_2$) determined in step 1b;

4b – If the difference determined in step 3b is greater than 0.25 %, add 1 mΩ/K to the value of $k$ and repeat steps 2b and 3b until at least the specified difference is reached. If the difference between $P_m$ and $P_2$ increases with increasing $k$, reduce $k$ in steps of 1 mΩ/K. When the specified power difference is reached, the value of $k$ is determined and the I-V curve is corrected to the desired conditions $G_2$ and $T_2$.

The flowchart of the developed procedure is shown in Fig. 1.

If the I-V curve is corrected for an irradiance higher than the measured value ($G_2 > G_1$), it is necessary to extend the measured I-V curve to complete the points of the curve until there is at least one point at $V_{oc}$ under the conditions $G_2$, $T_2$. The linear extrapolation, i.e. the extension by a line obtained by linear regression of the points from $I = 0$ (or the lowest measured current) up to 20 % of the maximum power current ($I_{mp}$), is a solution for small irradiance corrections estimated up to 10 %. For higher corrections, e.g. from $G_1 = 700$ W/m$^2$ to $G_2 = 1000$ W/m$^2$, the linear extrapolation tends to overestimate $V_{oc}$ of the corrected I-V curve. To minimize this effect, the extrapolation in the $V_{oc}$ range (from the first to the fourth quadrant) is performed using an exponential model calculated from the analytical parameters of the measured I-V curve [27]. In this way, the added points retain the exponential trend of the I-V curve and avoid the distortion that occurs when using a simple linear extrapolation.

The $I_{sc}$ region also has extrapolated points when the target temperature $T_2$ is lower than the measured temperature $T_1$. The calculation of $I_{sc}$ and the extrapolation of the I-V curve to the second quadrant in the case of temperature variations is performed using a linear regression line calculated from the points corresponding to $V = 0$ or the lowest measured voltage, up to 20 % of the maximum power voltage ($V_{mp}$).

The correction method was tested for accuracy when applied to a series of I-V curves obtained under controlled indoor conditions. After validation, it was applied to an outdoor PV string consisting of five PV modules connected in series. All the necessary procedures to obtain the translated curve were implemented in a spreadsheet using the Visual Basic for Applications (VBA) programming language.

### 2.2. Accuracy verification of the method for indoor I-V curves

For the analysis of the developed method by indoor measurements, eight crystalline silicon modules were characterized on a solar simulator at different values of $G$ and $T_c$, resulting in matrices of I-V curves. The I-V curves were measured at 500, 600, 700, 800, 900 and 1000 W/m$^2$ at temperatures of 25, 35, 45, 55 and 65 °C. Details of the devices and the methodology used in the measurements are described in another article [28]. The nominal characteristics of the modules used in this work are summarized in Table 1. The samples are a good selection of different monocrystalline (m-Si) and polycrystalline (p-Si) silicon modules with different efficiencies and cell manufacturing technologies.

The I-V curve under different conditions is corrected to STC according to the exact steps of procedure 1 described in IEC 60891. The same I-V curves are corrected to STC using the proposed method. The results of both methods are compared using Eq. (4), i.e. the difference

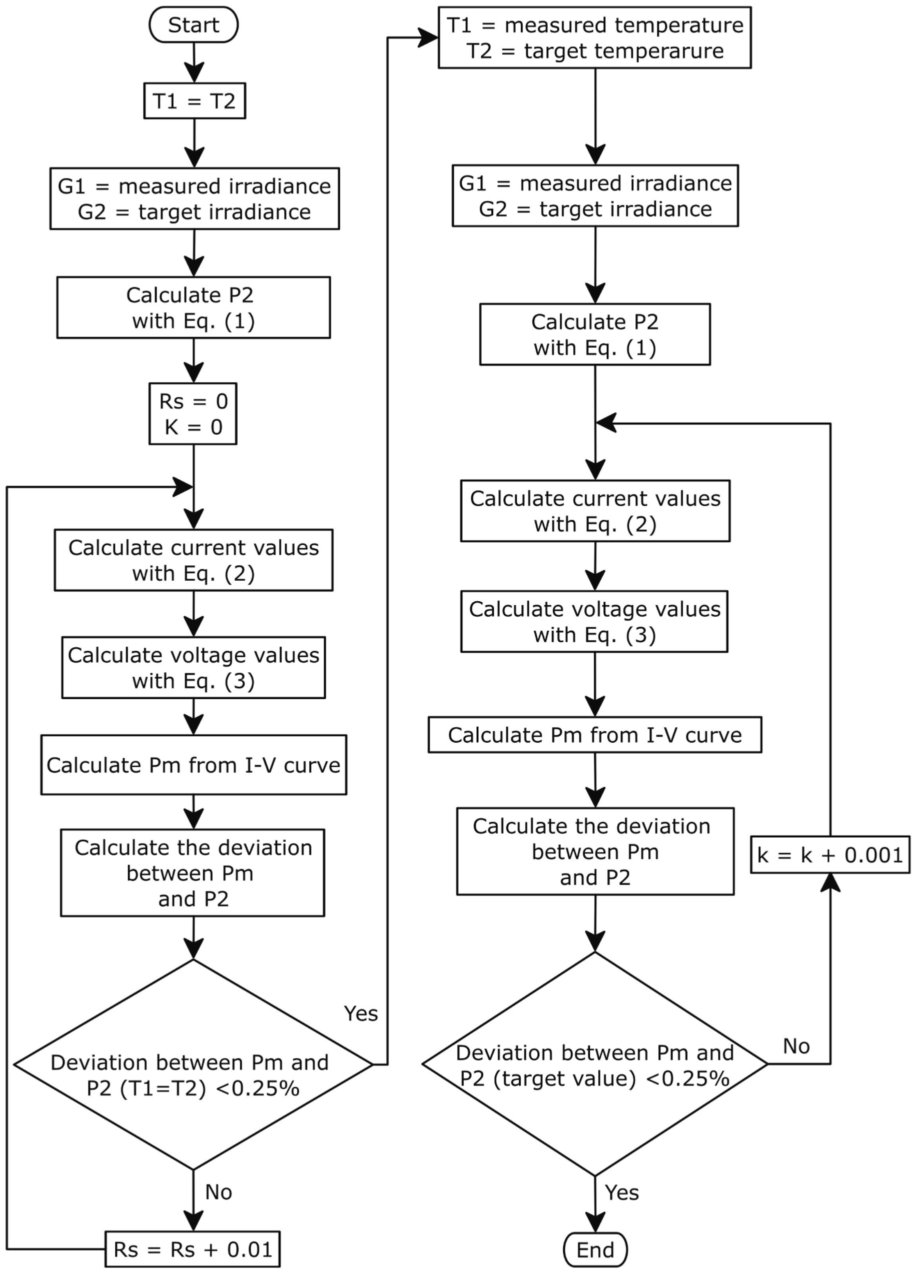


**Fig. 1.** Flowchart of the developed procedure for I-V curve correction.

**Table 1**
Nominal characteristics of the PV modules used in the indoor tests.

| Module | Cell Material | Number of Cells | $P_m$ (W) | $I_{sc}$ (A) | $V_{oc}$ (V) | $\eta$ (%) | α (%/°C) | β (%/°C) | γ (%/°C) |
|---|---|---|---|---|---|---|---|---|---|
| A | m-Si | 60 | 290 | 9.80 | 39.30 | 17.60 | 0.05 | −0.30 | −0.43 |
| B | m-Si | 60 | 260 | 8.99 | 37.80 | 16.16 | 0.06 | −0.35 | −0.45 |
| C | m-Si | 60 | 265 | 9.11 | 37.90 | 16.47 | 0.06 | −0.35 | −0.45 |
| D | m-Si PERC | 144 (half cell) | 400 | 10.36 | 48.93 | 19.90 | 0.06 | −0.30 | −0.39 |
| E | p-Si | 72 | 315 | 9.01 | 46.20 | 16.23 | 0.06 | −0.31 | −0.41 |
| F | m-Si PERC | 144 (half cell) | 450 | 11.58 | 49.6 | 20.70 | 0.05 | −0.27 | −0.37 |
| G | m-Si PERC | 144 (half cell) | 400 | 10.42 | 49.39 | 19.86 | 0.05 | −0.32 | −0.42 |
| H | p-Si | 60 | 245 | 8.63 | 37.80 | 15.00 | 0.05 | −0.28 | −0.35 |

between the parameters obtained from the corrected curve ($I_{sc}$, $V_{oc}$ and $P_m$) and the measured curve under STC.

$$\Delta(\%) = \frac{X - X_{STC}}{X_{STC}} \times 100 \tag{4}$$

where Δ(%) is the percentage deviation of the parameter from the value at STC, X is the generic evaluated parameter and $X_{STC}$ is the generic parameter at STC.

### 2.3. Application of the method to an outdoor PV string

A series of five PV modules of model H (245 W) described in Table 1 was installed outdoors to measure the I-V curve on a clear sky day. The measured and corrected I-V curves were compared with a reference measurement from the same day, that was as close as possible to the STC. To obtain this reference measurement, the modules in the string were covered with opaque material until they reached thermal equilibrium with the ambient temperature. All curves of the day are corrected to this reference curve so that the method can be checked for accuracy.

The modules were oriented at an inclination of 30° to true north, as the measurements were taken in the city of Porto Alegre, Brazil (30° S, 51° W). A pyranometer, an m-Si reference cell and a reference PV module of the same model connected to a shunt for $I_{sc}$ measurement were installed in the same plane. In addition, a T-type thermocouple was attached to each module (including the reference module) to measure $T_c$.

Data from the pyranometer, reference module, and thermocouples were recorded by an Agilent Model 34972A data acquisition device. The reference module was installed in the same plane as the PV string and a 60 mV/10 A manganin shunt resistor was connected to it. The calibration constant for the reference module is 54.167 mV/(1000 W/m$^2$) measured in a PASAN Sunsim 3c solar simulator. Since the module is calibrated under STC, the effects of temperature on the irradiance measurement are compensated by the coefficient α with a value of 0.0005 °C$^{-1}$.

The I-V curve of the PV string is measured using the PVPM 1100C curve tracer from PVE Photovoltaik Engineering. This device enables the measurement of PV systems with voltage and current limits of 1000 V and 100 A respectively. It measures the irradiance using the NES SOZ-03 reference cell, which also has temperature compensation with a Pt1000 temperature sensor. The device measures the temperature of the PV modules using a Pt100 sensor.

The I-V curves of the PV string were measured every 10 min on a clear sky day. The curve tracer and the data acquisition device were synchronized in time to obtain the irradiance data of the reference module, the reference cell and the pyranometer as well as the temperatures of the individual modules at the same time of curve acquisition.

The use of PV modules of the same model as the modules that make up the string enables coherent measurement of the effective irradiance at the string plane, as specified in the IEC 62446–1 standard. This module has the same spectral and angular response as the modules in the system, reducing the mismatch between the response of the string and the reference module. Fig. 2 shows the string used in the outdoor measurements.

## 3. Results and Discussions

The results were divided into indoor and outdoor tests. The indoor results evaluate the proposed method compared to the application of Procedure 1 of IEC 60891. The outdoor results estimate the differences in the translated parameters to a reference curve measured at midday, as close as possible to the STC.

### 3.1. Results of indoor tests

Part of the data analysis for the indoor tests consisted of evaluating the distribution of deviations for the most important parameters of the I − V curve. These deviations were calculated between the curve measured at STC and the curves corrected to STC. The deviations were analyzed separately for the correction according to the IEC 60891 standard and the proposed method. Fig. 3 shows the distribution plots for the deviations in $P_m$, $I_{sc}$ and $V_{oc}$ for both the IEC 60891 standard and the proposed method. The results include all modules described in Table 1 with $G$ values from 500 to 1000 W/m$^2$ and $T_c$ values from 25 to 65 °C, resulting in a total of 29 corrections per tested module. Taking into account 8 modules tested, of which three measurements were discarded, the total number of translations was 229. This sample included crystalline silicon modules from different manufacturing years and technologies, allowing a comprehensive assessment of the applicability of the method to crystalline silicon modules.

From the graphs in Fig. 3a and 3b, it can be deduced that for both translation methods, over 70 % of the corrected $P_m$ show deviations of ± 1 % compared to STC. The IEC 60891 method showed an absolute deviation of more than 2.5 % in five cases, while the highest absolute deviation of the proposed method was 2.5 %. It can also be observed that the IEC method has a slight tendency to underestimate the corrected $P_m$ value, while the proposed method tends to overestimate it. The distributions of $I_{sc}$ deviations shown in Fig. 3c and 3d are practically identical between the methods since the equations are the same. As for the $V_{oc}$ values, the proposed method provides a similar dispersion of deviations compared to the IEC 60891 method.

The results shown in Fig. 3 show that both methods lead to similar results, with the proposed method being even better for $V_{oc}$. It also has the advantage that no additional measurements of the I-V curve are

**Fig. 2.** PV string used for outdoor measurements and to verify the proposed method. The third module from left to right is the reference module for measuring solar irradiance.

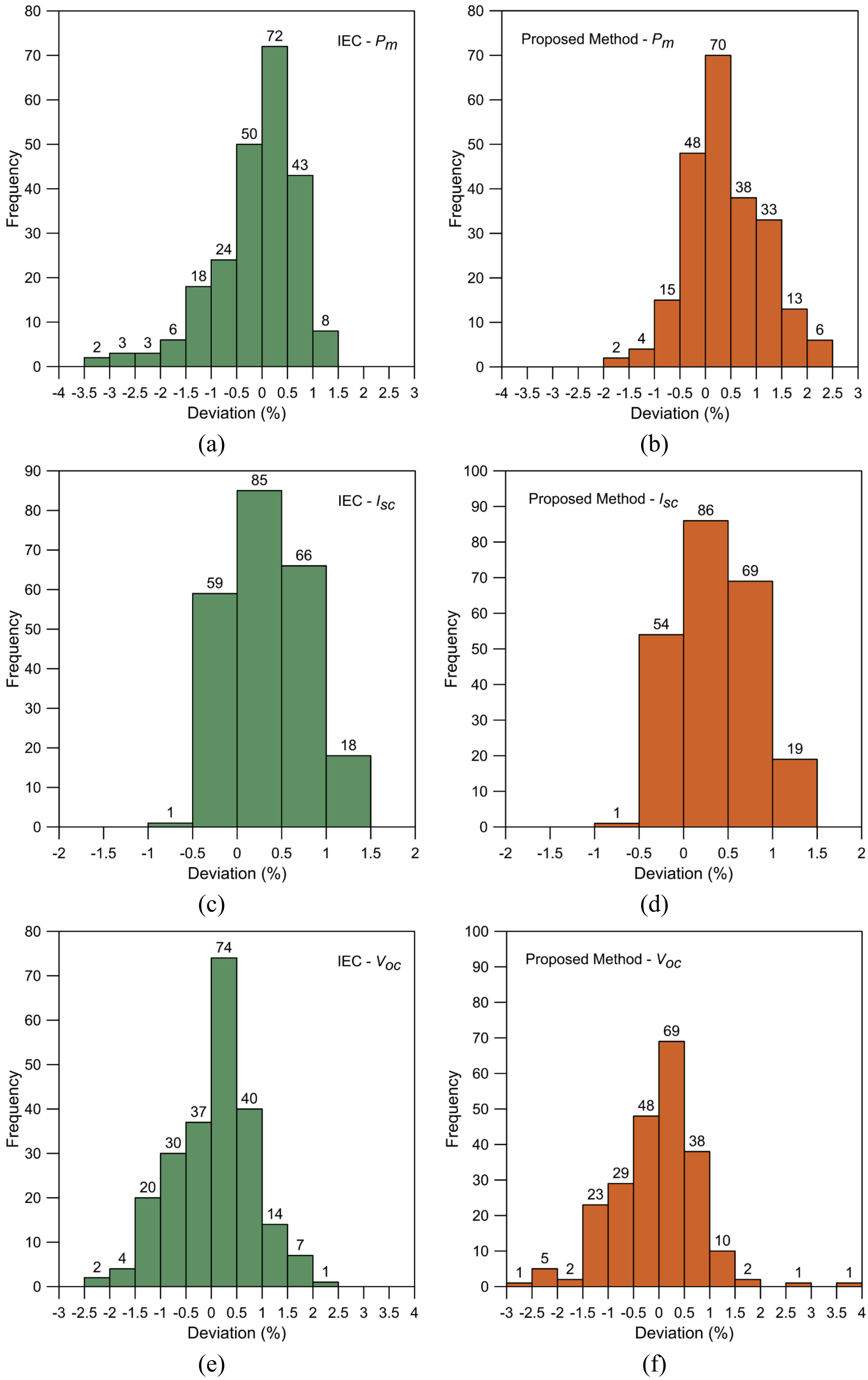


**Fig. 3.** The histograms in (a), (c) and (e) show the distribution of the deviations of the parameters $P_m$, $I_{sc}$ and $V_{oc}$ of the corrected curves compared to the curve measured at STC with the IEC 60891 method. The histograms in (b), (d) and (f) show the deviations of the parameters relative to STC using the proposed method.

required. The translation is based solely on the measured curve and only the three temperature coefficients are needed for the calculation, values that are readily available in the manufacturer's catalogs.

Further analysis of the proposed method included the evaluation of the mean (μ) and standard deviation (σ) of the electrical parameters ($P_m$, $I_{sc}$ and $V_{oc}$) and the deviation (Δ) between the mean and the STC value. The values obtained for each of the tested modules using the IEC 60891 method are shown in Table 2 and the results for the proposed method are shown in Table 3.

The results show the equivalence of the two methods of I-V curve translation in terms of deviations. The low values of the standard deviation indicate a low dispersion around the mean in the individual analysis of the modules. In this respect, the comparison of the mean electrical parameters with the values at STC allows an evaluation of the accuracy of the proposed method, as shown in Table 3. In the IEC 60891 method, all absolute deviations remained below 1 %, while in the proposed method, only module A showed an absolute deviation of more than 1 % (1.36 %) for $P_m$. This is because the proposed method assumes that $P_m$ varies linearly with $G$ and $T_c$. This assumption is reasonable if there is no significant variation of module efficiency with irradiance in the evaluated range, which is not the case for module A in particular.

Table 4 shows the percentage of corrections in the I-V curve to the total number of tests for $P_m$, $I_{sc}$ and $V_{oc}$, which are within ± 0.5 %, ±1.0 % and ± 1.5 % of the reference STC measurement. The proposed method has an uncertainty of ± 1.5 % for $P_m$ with 91 % coverage, while for $I_{sc}$ and $V_{oc}$ 100 % of the results were within ± 1.5 %.

The assessment of expected uncertainties in applying IEC60891 Procedure 1 and the proposed method is conducted through statistical methods. Table 5 displays symmetrical deviations in the electrical parameters for individual sample modules (with 29 corrections for each module) and a cumulative total of 229 corrections, covering 95 % of the translations. In essence, these values signify the uncertainty estimated through statistical methods for correcting the I-V curve, ensuring a coverage factor of 95 % of the conducted measurements.

With a coverage of 95 % for the whole sample (229 measurements), it can be noted that the uncertainties of the correction method are practically identical and below 2 %, both for IEC60891 procedure 1 (1.75 % for $P_m$) and for the proposed method (1.79 % for $P_m$). It is important to emphasize that the proposed method is performed without additional measurements. In both methods, there were cases of specific modules where the uncertainty value for the parameter $P_m$ exceeded 2 %. In the proposed method, this event occurred for module A (2.27 %) while in the IEC60891 method, it occurred for module C (3.10 %). In addition, both methods for module E exceeded the 2 % for $V_{oc}$.

The evaluation of the consistency of the $R_s$ and $k$ parameters for the proposed method compared to the IEC 60891 was performed by calculating the mean and standard deviation for 29 sets of $R_s$ and $k$ values for each module, as shown in Table 6. From the results, it can be inferred that in the proposed method, the mean of the $R_s$ values is consistent with the values obtained by the IEC when one standard deviation is considered, except for modules A and H, which require more than one standard deviation to be covered. Equivalence is achieved for the $k$ values within one standard deviation.

### 3.2. Outdoor test results

The results of the outdoor PV string tests were evaluated in the same way as the indoor tests, with the exception that no correction was made following procedure 1 of IEC60891. Since different instruments can be used to measure $G$ in the outdoor string measurements, including a pyranometer, a reference cell and a calibrated PV reference module, an individual analysis of the results was performed for each radiation sensor.

Fig. 4 shows the $G$ values of the three sensors and the average $T_c$ value of the string over one day. Between 11:12 and 12:24, data collection was interrupted as the string was covered to lower the temperature and bring the reference I-V curve as close as possible to the STC value. This happened at 12:24 when the average $T_c$ was 33 °C and $G$ = 1015 W/m$^2$. The I-V curve, which is used as a reference for evaluating the translation method applied to the PV string, does not correspond exactly to the STC value. However, since the objective is to evaluate the accuracy of the method, it was considered sufficient to use a condition close to the STC as a reference.

Throughout the entire test day, 33 I-V curves were recorded for the string. Fig. 5 shows histograms of the deviations of $P_m$ (a), $I_{sc}$ (b) and $V_{oc}$ (c) using the calibrated module as the irradiance sensor. Fig. 6 refers to the pyranometer and Fig. 7 to the reference cell as the irradiance sensor.

The use of different irradiance sensors leads to deviations, despite all sensors being calibrated and providing equivalent results near solar noon. These differences are related to spectral mismatch and non-uniform radiation between the PV modules and the sensors, especially when using the reference cell or the pyranometer. In the case of the reference module, which is identical to the modules of the string, the main difference lies in the spatial distribution of the radiation, as each module has a different view factor and some are affected differently by the radiation reflected from the surroundings.

Table 7 shows the percentage coverage for $P_m$, $I_{sc}$ and $V_{oc}$, which is within ± 0.5 %, ±1.0 % and ± 1.5 % of the reference values for three irradiance sensors. Table 8 shows the mean value, standard deviation and the difference between the mean value and the reference value of the electrical parameters of the PV string.

From the data in Tables 7 and 8, it can be deduced that the use of a reference module to measure irradiance leads to better results for $P_m$. As already mentioned, this is due to the better spectral match and the same cosine response between the reference module and the string modules. The worst results are obtained when using the reference cell, which has the highest absolute deviation. In this respect, it is plausible to believe that the use of a reference module significantly reduces the errors caused by the irradiance measurement, leaving only the errors associated with the temperature inherent to the correction method.

To evaluate the uncertainty of the method at 95 % coverage, Table 9 shows the deviations of the electrical parameters of the string for 95 % of the 33 corrections during the day. It can be seen that the proposed method has better precision in this case when the pyranometer is used as

**Table 2**
Mean value (μ), standard deviation (σ) and percentage deviation (Δ%) of the electrical parameters for each module considering the curves translated to STC according to the IEC 60891 procedure.

| Module | $P_m$ (W) | | | | $I_{sc}$ (A) | | | | $V_{oc}$ (V) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | STC | μ | σ | Δ(%) | STC | μ | σ | Δ(%) | STC | μ | σ | Δ(%) |
| A | 286.6 | 287.6 | 1.5 | 0.34 | 9.85 | 9.91 | 0.05 | 0.59 | 39.39 | 39.28 | 0.22 | −0.29 |
| B | 261.3 | 259.8 | 1.8 | −0.59 | 9.05 | 9.06 | 0.03 | 0.11 | 37.63 | 37.65 | 0.24 | 0.05 |
| C | 263.1 | 260.6 | 3.1 | −0.96 | 9.11 | 9.15 | 0.04 | 0.49 | 37.72 | 37.72 | 0.30 | 0.00 |
| D | 401.3 | 402.4 | 2.3 | 0.26 | 10.37 | 10.40 | 0.04 | 0.29 | 48.35 | 48.70 | 0.20 | 0.71 |
| E | 318.4 | 317.9 | 1.8 | −0.16 | 9.07 | 9.08 | 0.03 | 0.07 | 45.36 | 45.22 | 0.45 | −0.31 |
| F | 452.9 | 453.1 | 3.1 | 0.05 | 11.54 | 11.58 | 0.05 | 0.34 | 49.48 | 49.64 | 0.29 | 0.32 |
| G | 395.1 | 396.4 | 2.4 | 0.34 | 10.24 | 10.29 | 0.05 | 0.51 | 48.63 | 48.78 | 0.39 | 0.33 |
| H | 240.9 | 240.7 | 1.5 | −0.07 | 8.77 | 8.80 | 0.04 | 0.36 | 37.40 | 37.27 | 0.26 | −0.33 |

**Table 3**
Mean value (μ), standard deviation (σ) and percentage deviation (Δ%) of the electrical parameters for each module considering the curves translated to STC using the proposed method.

| Module | $P_m$ (W) | | | | $I_{sc}$ (A) | | | | $V_{oc}$ (V) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | STC | μ | σ | Δ(%) | STC | μ | σ | Δ(%) | STC | μ | σ | Δ(%) |
| A | 286.6 | 290.5 | 1.8 | 1.36 | 9.85 | 9.91 | 0.05 | 0.59 | 39.39 | 39.45 | 0.30 | 0.16 |
| B | 261.3 | 262.4 | 1.1 | 0.40 | 9.05 | 9.08 | 0.03 | 0.33 | 37.63 | 37.60 | 0.23 | −0.07 |
| C | 263.1 | 263.8 | 0.8 | 0.25 | 9.11 | 9.16 | 0.04 | 0.52 | 37.72 | 37.68 | 0.30 | −0.12 |
| D | 401.3 | 404.0 | 3.1 | 0.67 | 10.37 | 10.40 | 0.04 | 0.29 | 48.35 | 48.44 | 0.17 | 0.18 |
| E | 318.4 | 316.6 | 1.7 | −0.55 | 9.07 | 9.08 | 0.03 | 0.07 | 45.36 | 45.06 | 0.47 | −0.67 |
| F | 452.9 | 452.6 | 1.4 | −0.06 | 11.54 | 11.58 | 0.05 | 0.34 | 49.48 | 49.47 | 0.25 | −0.03 |
| G | 395.1 | 397.3 | 2.2 | 0.58 | 10.24 | 10.29 | 0.05 | 0.51 | 48.63 | 48.53 | 0.32 | −0.20 |
| H | 240.9 | 242.4 | 1.5 | 0.60 | 8.77 | 8.80 | 0.04 | 0.36 | 37.40 | 37.46 | 0.34 | 0.17 |

**Table 4**
Percentage of corrections in the I-V curve for $P_m$, $I_{sc}$ and $V_{oc}$, with results within ± 0.5 %, ±1.0 % and ± 1.5 % of the STC value for both methods evaluated.

| Method | $P_m$ | | | $I_{sc}$ | | | $V_{oc}$ | | |
|---|---|---|---|---|---|---|---|---|---|
| | ±0.5 % | ±1.0 % | ±1.5 % | ±0.5 % | ±1.0 % | ±1.5 % | ±0.5 % | ±1.0 % | ±1.5 % |
| Proc. 1 IEC 60891 | 53 % | 82 % | 94 % | 63 % | 92 % | 100 % | 48 % | 79 % | 94 % |
| Proposed | 52 % | 75 % | 91 % | 61 % | 92 % | 100 % | 51 % | 80 % | 95 % |

**Table 5**
Values of deviations in electrical parameters for 95% coverage for each module tested indoors and for the whole set of measurements.

| Module | $P_m$ | | $I_{sc}$ | | $V_{oc}$ | |
|---|---|---|---|---|---|---|
| | IEC | Proposed | IEC | Proposed | IEC | Proposed |
| A | ± 1.05 % | ± 2.27 % | ± 1.10 % | ± 1.13 % | ± 1.16 % | ± 1.50 % |
| B | ± 1.86 % | ± 1.10 % | ± 0.85 % | ± 0.98 % | ± 1.25 % | ± 1.13 % |
| C | ± 3.10 % | ± 0.64 % | ± 1.28 % | ± 1.29 % | ± 1.27 % | ± 1.34 % |
| D | ± 1.13 % | ± 1.74 % | ± 0.93 % | ± 0.92 % | ± 1.37 % | ± 0.63 % |
| E | ± 1.43 % | ± 1.79 % | ± 0.63 % | ± 0.74 % | ± 2.01 % | ± 2.09 % |
| F | ± 1.36 % | ± 0.67 % | ± 0.82 % | ± 1.12 % | ± 1.53 % | ± 1.03 % |
| G | ± 0.78 % | ± 1.58 % | ± 0.82 % | ± 1.06 % | ± 0.48 % | ± 1.23 % |
| H | ± 1.30 % | ± 1.45 % | ± 1.02 % | ± 1.05 % | ± 1.53 % | ± 1.57 % |
| Total | ± 1.75 % | ± 1.79 % | ± 1.09 % | ± 1.11 % | ± 1.53 % | ± 1.57 % |

**Table 6**
$R_s$ e $k$ according to IEC 60891, mean value for $R_s$ and $k$ from the proposed method and standard deviation (σ) for $R_s$ and $k$ from the proposed method for eight PV modules.

| Module | IEC 60891 | | Proposed method | | | |
|---|---|---|---|---|---|---|
| | $R_s$ (Ω) | $k$ (mΩ/K) | $R_s$ (Ω) | σ (Ω) | $k$ (mΩ/K) | σ (mΩ/K) |
| A | 0.34 | 3 | 0.22 | 0.09 | 3 | 2 |
| B | 0.23 | 2 | 0.19 | 0.09 | 3 | 2 |
| C | 0.24 | 1 | 0.18 | 0.11 | 3 | 2 |
| D | 0.13 | 1 | 0.18 | 0.08 | 2 | 1 |
| E | 0.21 | 4 | 0.22 | 0.11 | 3 | 2 |
| F | 0.16 | 2 | 0.17 | 0.08 | 2 | 1 |
| G | 0.13 | 2 | 0.18 | 0.09 | 3 | 2 |
| H | 0.39 | 3 | 0.21 | 0.13 | 2 | 1 |

a radiation sensor, as there were measurements with the reference module that had greater deviations in the extremes of the day. For Pm, there were deviations of ± 1.83 % with the pyranometer and about ± 2.10 % (2.09 %) with the module.

It is important to emphasize that there are other variables and sources of uncertainty in outdoor measurements. However, when using a pyranometer or a calibrated PV module, the new method presented in this study should achieve an uncertainty of about ± 2 % for $P_m$ and $I_{sc}$ and 1 % for $V_{oc}$. This uncertainty is slightly higher than the results obtained indoors, where conditions can be better controlled, but is still within a typical range of ± 2 % at 95 % coverage. This uncertainty range takes into account the measurements during a whole day under clear skies, starting at 9 am and ending at 3:30 pm, which is indeed a plausible time frame for outdoor measurements.

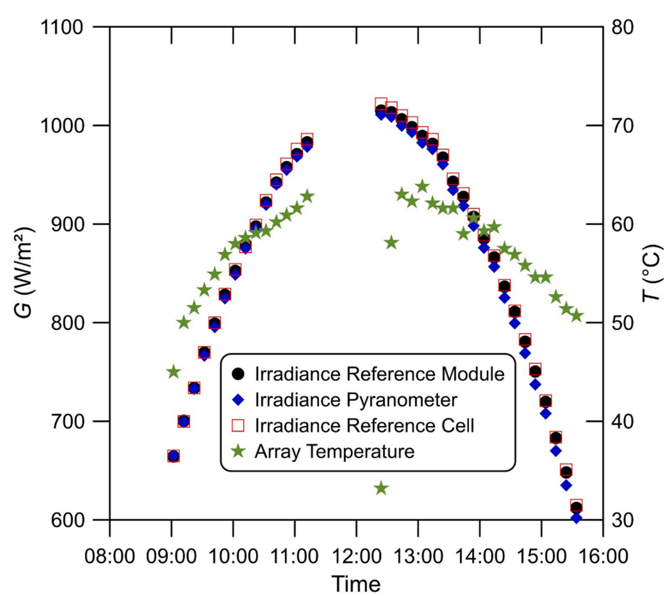


**Fig. 4.** Irradiance of the three sensors and the representative temperature of the PV string for each I–V curve over time.

## 4. Conclusions

The correction of the I-V characteristic curve to STC is of crucial importance for the electrical characterization of photovoltaic modules and strings, especially for outdoor measurements. The procedures of the

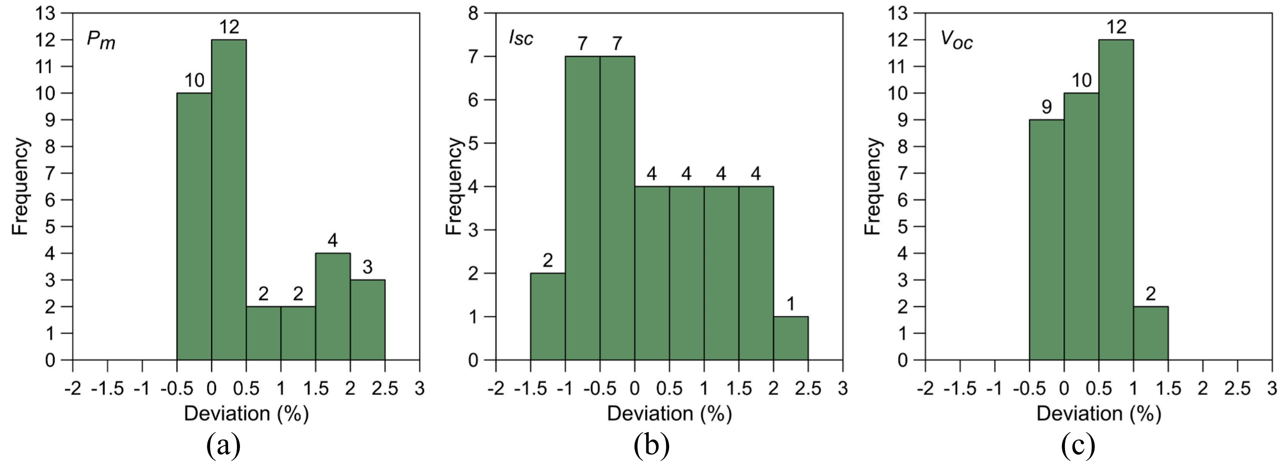


**Fig. 5.** Histograms showing the distribution of deviations for $P_m$ (a), $I_{sc}$ (b) and $V_{oc}$ (c) using the calibrated module as irradiance sensor.

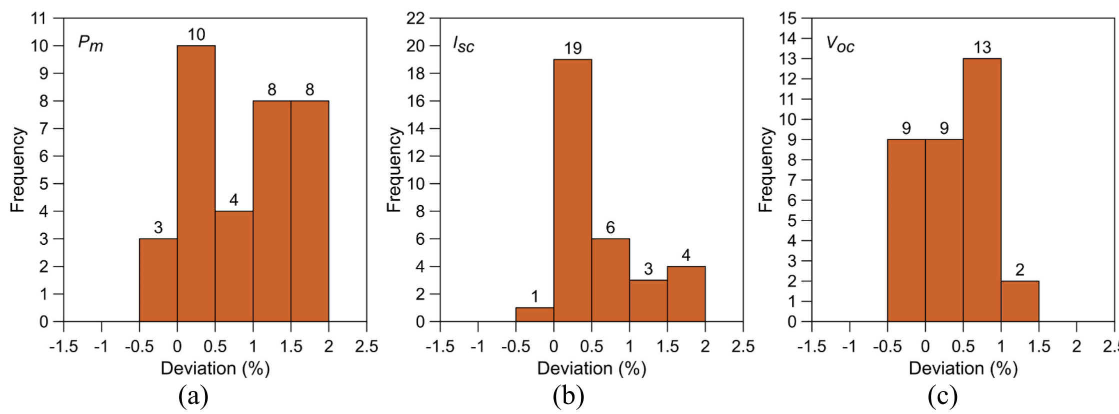


**Fig. 6.** Histograms showing the distribution of deviations for $P_m$ (a), $I_{sc}$ (b) and $V_{oc}$ (c) using the pyranometer as irradiance sensor.

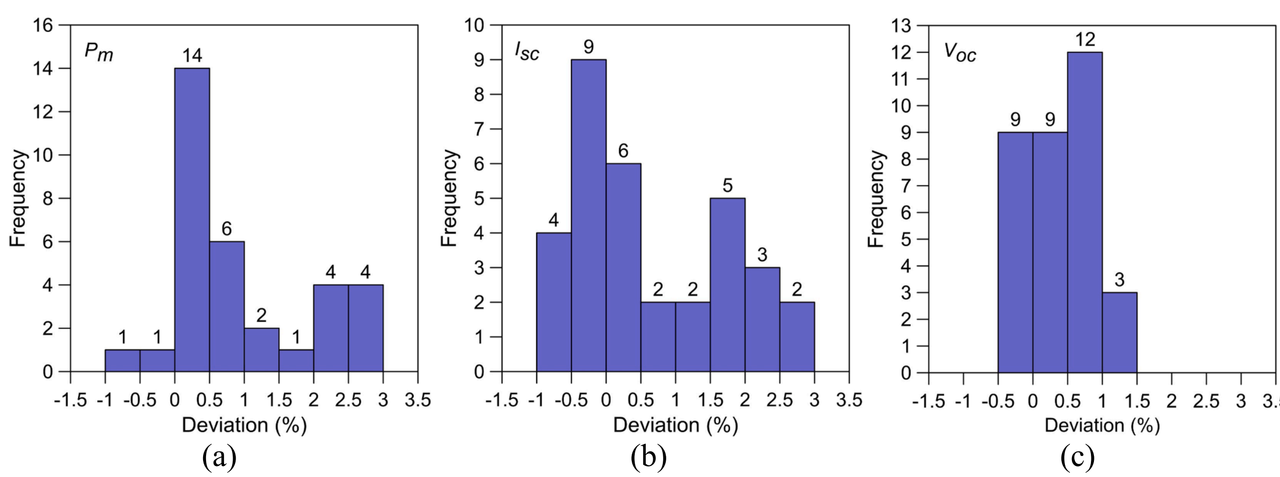


**Fig. 7.** Histograms showing the distribution of deviations for $P_m$ (a), $I_{sc}$ (b) and $V_{oc}$ (c) using the reference cell as irradiance sensor.

**Table 7**
Percentage of corrections in the I-V curve for $P_m$, $I_{sc}$ and $V_{oc}$, with results within ± 0.5 %, ±1.0 % and ± 1.5 % of the STC value for three irradiance sensors.

| RadiationSensor | $P_m$ | | | $I_{sc}$ | | | $V_{oc}$ | | |
|---|---|---|---|---|---|---|---|---|---|
| | ±0.5 % | ±1.0 % | ±1.5 % | ±0.5 % | ±1.0 % | ±1.5 % | ±0.5 % | ±1.0 % | ±1.5 % |
| Module | 67 % | 73 % | 79 % | 33 % | 67 % | 79 % | 58 % | 94 % | 100 % |
| Pyranometer | 39 % | 52 % | 76 % | 61 % | 79 % | 88 % | 54 % | 94 % | 100 % |
| Cell | 45 % | 64 % | 70 % | 45 % | 52 % | 58 % | 54 % | 91 % | 100 % |

**Table 8**
Mean value (μ), standard deviation (σ) and percentage deviation (Δ%) of the electrical parameters of the PV string for three irradiance sensors.

| Radiation Sensor | $P_m$ (W) | | | | $I_{sc}$ (A) | | | | $V_{oc}$ (V) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Ref | μ | σ | Δ(%) | Ref | μ | σ | Δ(%) | Ref | μ | σ | Δ(%) |
| Module | 1144 | 1150 | 9.2 | 0.54 | 8.91 | 8.93 | 0.09 | 0.25 | 180.69 | 181.37 | 0.82 | 0.37 |
| Pyranometer | 1144 | 1154 | 8.0 | 0.90 | 8.91 | 8.96 | 0.05 | 0.61 | 180.69 | 181.43 | 0.85 | 0.41 |
| Cell | 1144 | 1155 | 11 | 0.94 | 8.91 | 8.97 | 0.10 | 0.65 | 180.69 | 181.44 | 0.84 | 0.41 |

**Table 9**
Values of the deviations in the electrical parameters for 95% coverage of outdoor measurements.

| Radiation Sensor | $P_m$ | $I_{sc}$ | $V_{oc}$ |
|---|---|---|---|
| PV Module | ± 2.09 % | ± 1.83 % | ± 0.99 % |
| Pyranometer | ± 1.83 % | ± 1.56 % | ± 0.99 % |
| Reference Cell | ± 2.76 % | ± 2.49 % | ± 1.16 % |

IEC 60891 standard require measurements at different irradiance levels and cell temperatures. This can be a complex process, especially for outdoor PV strings or arrays where such measurements are impractical.

In this study, a simplified method for correcting the I-V curve based solely on the measured curve and temperature coefficients was presented. It was evaluated for an irradiance of 500 to 1000 W/m$^2$ and a temperature of 25 to 65 °C. With this new correction method, it is possible to correct an I-V curve without determining correction coefficients, as described in the IEC 60891 standard.

In the indoor tests, the deviation between the corrected value and STC was ± 1.79 % for $P_m$, ±1.11 % for $I_{sc}$ and ± 1.57 % for $V_{oc}$, taking into account all 229 measurements and a coverage of 95 %. These values were practically identical to those determined using Procedure 1 of the IEC 60891 standard. It was found that Procedure 1 tends to slightly underestimate the $P_m$ of the translated curve, while the proposed method tends to slightly overestimate $P_m$.

Measurements on an outdoor PV string were also used to evaluate the performance of the proposed method. A total of 33 curves were measured on a clear sky day with different $G$ and $T_c$ values. All curves were translated to the measured value of 1015 W/m$^2$ and 33 °C, which occurred around noon and was used as a reference. With 95 % coverage, the deviation was ± 1.83 % for $P_m$, ±1.56 % for $I_{sc}$ and ± 0.99 % for $V_{oc}$, using a pyranometer as the irradiance sensor.

The developed method has an accuracy comparable to IEC 60891 procedure 1 and in some cases even better. It provides a reliable and simplified method for correcting I-V curves without the need for measurements under different conditions, which are necessary to calculate parameters such as $R_s$ and $k$. The implementation is relatively simple and can be made available in a spreadsheet. The method can also be implemented directly in I-V curve tracers, requiring only the values of the temperature coefficients as input, making it a straightforward process. Future work is planned to evaluate the proposed method for thin-film PV modules, and to implement the methodology and algorithms in a user-friendly application.

## CRediT authorship contribution statement

**Fernando Schuck de Oliveira:** Investigation, Methodology, Software, Validation, Writing – original draft, Writing – review & editing. **Fabiano Perin Gasparin:** Conceptualization, Investigation, Methodology, Project administration, Software, Validation, Writing – original draft, Writing – review & editing. **Felipe Detzel Kipper:** Investigation, Writing – review & editing. **Arno Krenzinger:** Supervision, Writing – review & editing.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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